KY002S MOSFET Switch Module Specification

- **Power Supply Voltage**: DC 3V–24V
- **Standby Current**: Less than 2 µA
- **Continuous Output Current**:
  - **5A** (when supply voltage is 5V–24V)
  - **4A** (when supply voltage is 4V–5V)
  - **2A** (when supply voltage is 3V–4V)
- **Trigger Type**: Falling edge trigger (via Tg pin)
- **Output State**: High level (open) when idle; output high upon trigger, then toggles on next trigger
- **Isolation**: **Optocoupler isolation** for enhanced noise immunity and safety
- **Pin Definitions**:
  - **VCC**: Power input
  - **VO**: Output terminal (connected to load)
  - **Tg**: Trigger signal input
  - **GND**: Ground
- **Size**: Compact, suitable for space-constrained applications
- **Application Note**: When driving inductive loads (e.g., relays, motors), connect a **freewheeling diode** in parallel with the load to protect the switching elements.

This module is designed for high-power, low-power-consumption applications, ideal for remote battery switching and IoT systems requiring reliable, long-term operation.

Brave AI-generated answer. Please verify critical facts.